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Enhanced piezoelectricity and excellent thermal stabilities in Nb–Mg co-doped $CaBi_4Ti_4O_{15}$ Aurivillius high Curie temperature ceramics

Lili Li[a,1], Hongbin Yuan[a,b,1], Peiming Huang[a], Yuhao Zhang[a], Linsheng Sheng[a], Peng Zheng[a], Lin Zhang[c], Fei Wen[a,*], Wei Wu[a], Zhuo Xu[a,c,**]

[a] Key Lab of RF Circuits and Systems of the Ministry of Education, College of Electronics and Information, Hangzhou Dianzi University, Hangzhou, 310018, China
[b] Qianjiang College, Hangzhou Normal University, Hangzhou, 310012, China
[c] Electronic Materials Research Laboratory, Key Laboratory of the Ministry of Education & International Center for Dielectric Research, School of Electronic Science and Engineering, Faculty of Electronic and Information Engineering, Xi'an Jiaotong University, Xi'an, 710049, China

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ABSTRACT

Nb–Mg co-doped $CaBi_4Ti_4O_{15}$ ceramics were synthesized for potential high temperature piezoelectric applications via the solid-state reaction technique. The influences of Nb–Mg doping level on structure, piezoelectricity and other electrical properties were systematically researched. The XRD, EDS and Raman scattering results confirmed that the doped Nb–Mg ions had successfully diffused into the crystal lattice of $CaBi_4Ti_4O_{15}$ ceramic. Co-doping of Nb–Mg enhanced the piezoelectric coefficient d_{33} effectively without causing significant deterioration on Curie temperature T_C. The $CaBi_4Ti_{3.9}(Nb_{2/3}Mg_{1/3})_{0.1}O_{15}$ ceramic possessed the optimized electric properties with a high d_{33} value of 24 pC/N accompanied by a high T_C value of 787 °C. In addition, this ceramic also exhibited a good thermal stability with d_{33} keeping 91% of its original value after annealing at 600 °C. All these results suggested that Nb–Mg co-doped $CaBi_4Ti_4O_{15}$ ceramic could be a potential candidate for high temperature piezoelectric sensor or detector applications.

1. Introduction

In recent years, there are growing demands for sensors that can be applied at high temperature, such as electric fuel injection piezoelectric valve for internal combustion, ultrasonic locator in nuclear reactor, *etc.* [1,2]. Hence, it is urgently needed to explore piezoelectric ceramics which has good piezoelectric properties with high Curie temperature and excellent thermal stabilities. Bismuth layer-structured ferroelectrics (BLSFs) ceramics have been considered as ideal candidates and caught great attention from both academics and industries. It is not only because their ultra-high Curie temperature (approximately 600–900 °C), but also the lead-free and low manufacturing cost [3,4].

The general formula of BLSFs is the $(Bi_2O_2)^{2+}(A_{m-1}B_mO_{3m+1})^{2-}$, of which $(Bi_2O_2)^{2+}$ is fluorite-like layer along the c axis, and unit of $(A_{m-1}B_mO_{3m+1})^{2-}$ is pseudo-perovskite structure [5,6]. In formula, A is a mono-, di-, trivalent ions (such as Na^+, Ca^{2+}, Sr^{2+}, La^{3+} and Bi^{3+}, *etc.*), or combination of them, as well as B is a tetravalent, pentavalent and hexavalent transition element (*e.g.*, Fe^{3+}, Cr^{3+}, Ti^{4+}, Nb^{5+}, Ta^{5+}, W^{6+}, Mo^{6+}) [7]. The range of (BO_6) octahedra m usually belongs to 1–6. $CaBi_4Ti_4O_{15}$ (CBT) is a typical Aurivillius structure ferroelectrics with $m = 4$, which A-site is consisted of Ca^{2+} and Bi^{3+} ions while B-site is occupied by Ti^{4+}. It was reported that pristine CBT had a rather high Curie point ($T_C = 790$ °C) and a low dielectric loss (tanδ = 0.2%), which made it as one of the most noteworthy high temperature piezoelectric materials [5,8]. However, the piezoelectric coefficient value ($d_{33} = 7$ pC/N) is extremely low because the switching of spontaneous polarization is limited to the a-b plane. Additionally, CBT possesses a low resistivity (ρ) and a large coercive field (E_c).

To improve the electrical properties of CBT-based ceramics, it would be a feasible approach to substitute isovalent or heterovalent ions for Ca^{2+}. Yan et al. obtained a high d_{33} value of 17 pC/N in CBT ceramic by Na and Ce doping at Ca-site [9]. Similar enhancements of d_{33} value were also achieved in K/Ce and Li/Ce doped CBT ceramics [10]. In addition, Sheng et al. recently obtained a CBT-based ceramics with a remarkable d_{33} value of 19 pC/N by introducing double rare earth ions Nd^{3+} and Ce^{3+} at Bi-site, which indicates that substitution of Bi^{3+} ions

* Corresponding author.
** Corresponding author. Key Lab of RF Circuits and Systems of the Ministry of Education, College of Electronics and Information, Hangzhou Dianzi University, Hangzhou, 310018, China.
E-mail addresses: wenfei@hdu.edu.cn (F. Wen), xuz2018@126.com (Z. Xu).
[1] These authors contributed equally: Lili Li and Hongbin Yuan.

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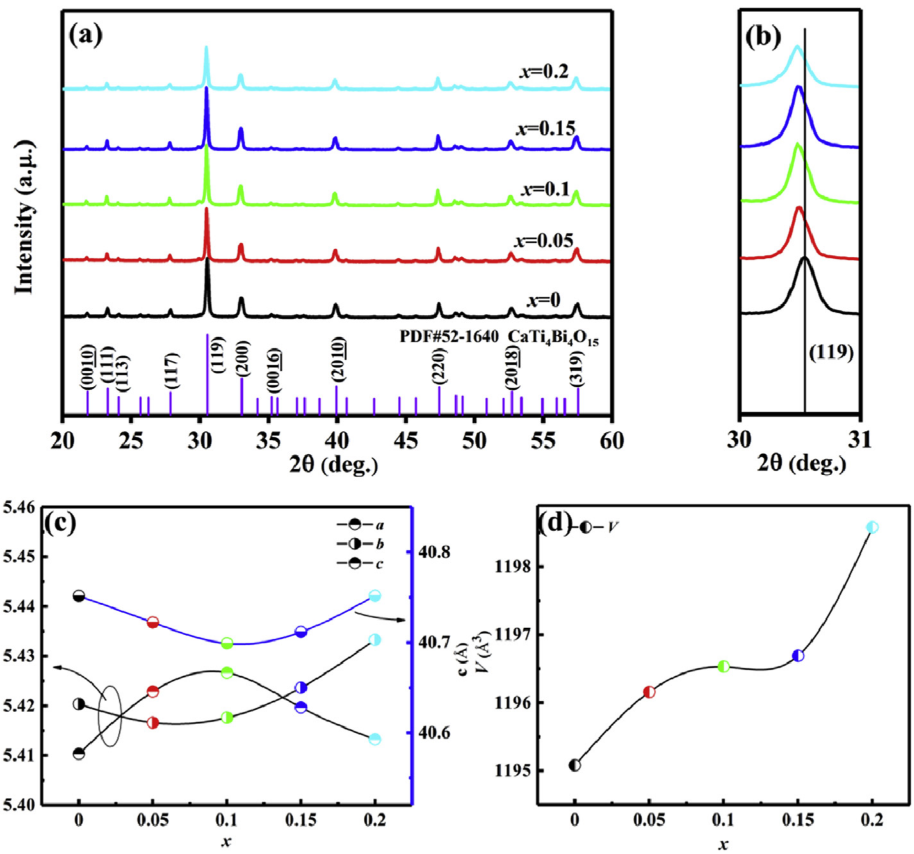

Fig. 1. (a) Room-temperature XRD spectra for CNM-x ceramics (x = 0, 0.05, 0.1, 0.15, 0.2), (b) the zoomed XRD patterns in the 2θ range of 30–31°, (c) lattice parameters a, b, and c, (d) cell volume for CNM-x ceramics.

with double rare earth ions is also a practicable way to improve the piezoelectric activities of CBT ceramic [11]. Nevertheless, the piezoelectric activities of ceramics reported by above literatures via double ions doping at Bi-site or Ca-site are still relatively low until now. In addition, the improvement of piezoelectricity of CBT ceramic by co-doping at Ca-site or Bi-site is usually accompanied by a deterioration of thermal stability, which greatly affects its practical applications at high temperature. Therefore, studies are still needed to fabricate CBT-based ceramics which have enhanced piezoelectric properties with optimized thermal stabilities.

In the past several decades, co-doping with mixed-valence ions at Ti-site is reported to be effective in strengthening piezoelectricity of the PZT system [12,13]. Some researchers recently had introduced this co-doping strategy into the modification of BLSFs. For example, Long et al. fabricated Nb–Mg co-doped $Bi_4Ti_3O_{12}$ ceramics with an excellent d_{33} value of 30 pC/N [14]. Li et al. prepared Nb–Cu co-doped the $Bi_4Ti_3O_{12}$ ceramics, and then achieved an increased d_{33} value of 38 pC/N [15]. Moreover, a remarkable d_{33} value of 15 pC/N was recently acquired in W/Cr co-doped $CaBi_2Nb_2O_9$ ceramics. All these work demonstrated the great potentials of co-doping with mixed-valence ions at Ti-site in improving the piezoelectric properties of BLSFs. Especially, Shen et al. recently introduced Nb and Mn ions to the Ti-site of $CaBi_4Ti_4O_{15}$ ceramics and an increased d_{33} value ~23 pC/N was obtained [16], which further implies that the co-doping strategy with mixed-valence ions at B-site is more potential to enhance the piezoelectricity of CBT-based ceramics. However, related studies with above method in CBT-based ceramic were rarely reported and the underlying enhancement mechanism was still not well elucidated.

In this study, Nb–Mg co-doped $CaBi_4Ti_4O_{15}$ ceramics are fabricated by conventional solid-state reaction method and the effects of co-doping on the structure and electrical properties of ceramics are systematically explored. Optimized piezoelectric properties with a high d_{33} value of 24 pC/N and a high T_C value of 787 °C along with a good thermal stability are achieved. Related mechanisms for the enhanced electrical performances are discussed rigorously.

2. Experimental

In this work, $CaBi_4Ti_{4\text{-}x}(Nb_{2/3}Mg_{1/3})_xO_{15}$ (abbreviated as CNM-x, x = 0, 0.05, 0.1, 0.15, 0.2) ceramics were synthesized through the conventional solid-state reaction. Analytical-purity oxides, Bi_2O_3(99.99%), $CaCO_3$(99.99%), MgO(99.9%), TiO_2(99.8%) and Nb_2O_5(99.5%), all used for raw materials, were weighted in accordance with the stoichiometric composition, and ball-milled for 12 h with alcohol in polyethylene bottles. The dried slurry was calcined for 4 h at 825 °C. And then the powders were milled again sequentially. Organic binder (PVA) was added to the mixtures to avoid crack and then pressed uniaxially into disks under a pressure of 150 MPa. After that, the disks were burned at 550 °C for 2 h in furnace to remove PVA. To reduce the volatility of Bi, the disks were surrounded by their own powders in the sealed alumina crucibles during sintering (975–1075 °C for 1 h). Finally, these disks, with thickness of 0.6–0.8 mm and diameter of 10–11.5 mm, were coated with high temperature silver paste and then heated for 2 h at 820 °C.

The crystal structures of the CNM-x ceramics were determined by the X-ray diffractometer (XRD, Ultima IV) with a CuK_α. The micro-morphology was observed with the scanning electron microscopy (SEM, JEOL JSM6380-LV). Raman spectra were collected by iHR320 using 532 nm laser line. Relative permittivity and dielectric loss against temperature were measured using impedance meter (Keysight E4990A). DC resistivity was recorded by the high-resistance meter (Keithley 2410) under 10 V. And the piezoelectric constant d_{33} of all

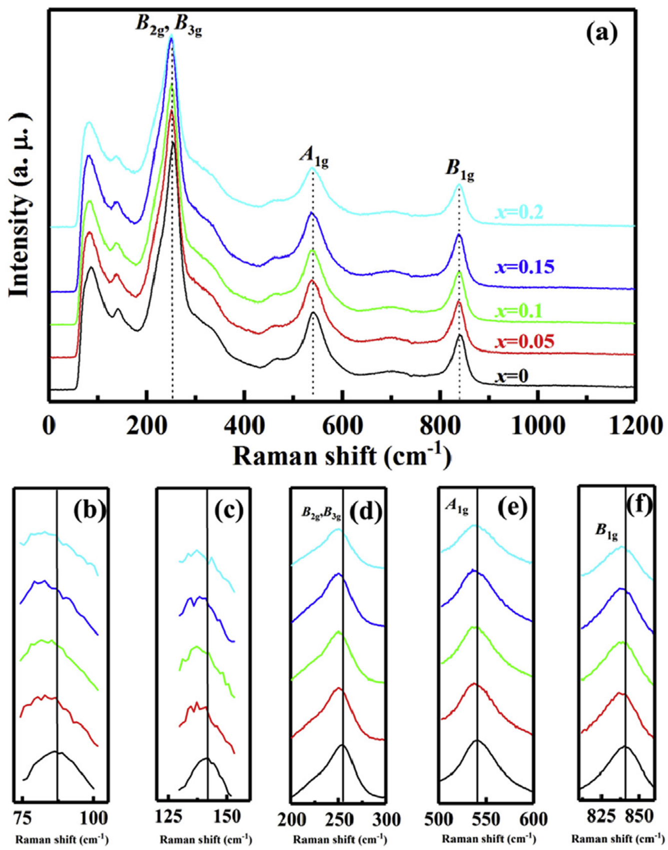

Fig. 2. (a) The Raman patterns of CNM-x at room temperature, (b)–(f) Raman peaks zoomed in different frequency.

samples were measured by a piezo-d_{33} m (ZJ-3AN).

3. Results and discussion

Fig. 1(a) displays the X-ray diffraction patterns of CNM-x ceramics within 2θ range of 20°-60°. As shown, all the XRD patterns present the typical characteristic of four-layered Aurivillius structure. The most intense peak of the compositions is indexed as (119), which agrees well with the conclusion of other works [17]. All samples form a single orthorhombic crystal structure without secondary phase, which suggested that doped Nb–Mg ions has favorably diffused into the crystal lattice of CBT. With an increase in Nb–Mg doping level, a slight left shift of (119) peak is observed in Fig. 1(b), indicating a slight lattice expansion. Detailed doping content dependence of the lattice parameters and cell volume are shown in Fig. 1 (c) and (d). With an increase in x, a continuous increase of cell volume can be observed, which is consistent with the left shift of (119) peak. Considering the fact that the ionic radius of Nb^{5+} (0.64 Å) and Mg^{2+} (0.72 Å) are larger than that of Ti^{4+} (0.605 Å) but smaller than that of Ca^{2+} (1.34 Å) and Bi^{3+} (1.17 Å), the substitutions of Ti^{4+} with Nb^{5+} and Mg^{2+} can be confirmed.

Fig. 2(a) displays the Raman spectra of CNM -x ceramics measured at room temperature in the spectral range 0–1200 cm^{-1}. Several sharp peaks including ~253, ~540 and ~839 cm^{-1} are observed in ceramics, which is similar to the result reported by He et al. [18,19]. It has been revealed that the phonon modes of bismuth layer-structured ferroelectrics could be split into two modes, one is high frequency mode (over 200 cm^{-1}) and the other is low frequency mode (below 200 cm^{-1}). The lower one is in connection with ion vibration at A-sites of $(Bi_2O_2)^{2+}$ and pseudo-perovskite layer, while the higher one has an association with (TiO6) octahedra structure [20,21]. Raman peaks zoomed in different frequencies are displayed in Fig. 2 (b)–(f). The "rigid layer" mode of ~86 cm^{-1} is associated with vibration of Bi^{3+} ion in $(Bi_2O_2)^{2+}$ structure, while the mode of ~142 cm^{-1} could be linked with vibration of A-sites of pseudo-perovskite layer, which is mainly influenced by the atom of Bi owing to relative atomic mass of Bi is heavier compared with that of Ca. The B_{2g} and B_{3g} modes (~253 cm^{-1}) is related to O–Ti–O vibration [22], while the sharp phonon mode A_{1g} (approximately at 540 cm^{-1}) is caused by relative displacement of oxygen atom in TiO_6 octahedron, and B_{1g} vibration mode (~839 cm^{-1}) is relevant to the stretching of the TiO_6 octahedron [18]. With increasing doping level, left shift could be observed for all the phonon modes, which further confirms the diffusion of doped Nb–Mg ions into the CBT crystal lattice.

Fig. 3 presents the surface morphologies of the Nb–Mg co-doped CBT ceramics. Obviously, plate-like grain structure could be observed for all ceramics, which is a typical characteristic of Aurivillius-structured ferroelectrics and could be ascribed to the high grain growth rate perpendicular to the c-axis [23]. The grain size shows no significant difference after doping, suggesting that the doped Nb–Mg ions has little impact on the grain growth of CBT ceramic. Still some pores can be observed in the Nb–Mg co-doped ceramics, all the samples exhibit a relative density higher than 92%. The elemental mappings of the untreated ceramics surface with x = 0.1 are given in Fig. 4. It is clear that each element displays a uniform distribution, which further confirms the diffusion of Nb/Mg ions in the pure CBT structure and in accordance with the results of XRD and Raman scattering.

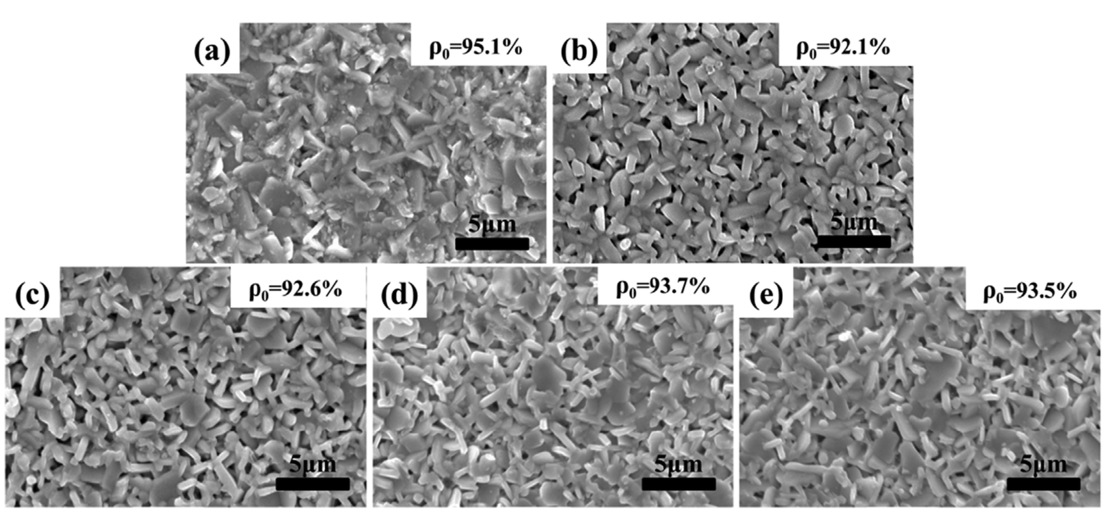

Fig. 3. SEM images of ceramics sample: (a) CBT, (b) CNM-0.05, (c) CNM-0.1, (d) CNM-0.15 and (e) CNM-0.2.

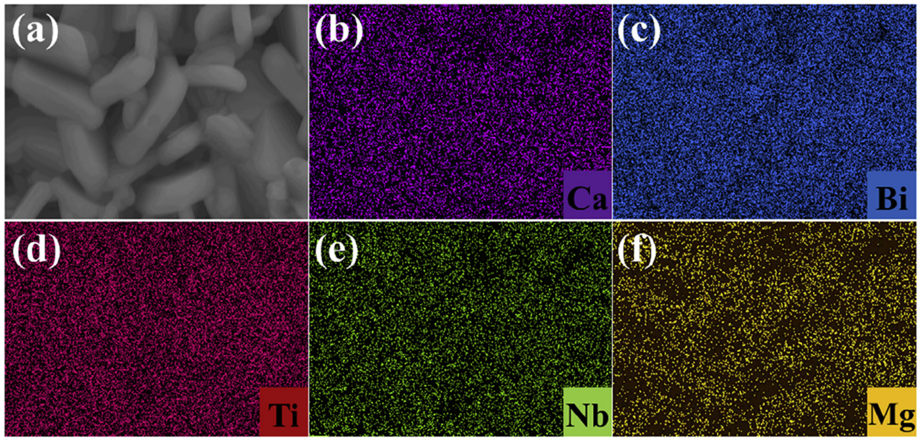

Fig. 4. (a) The SEM image of CNM-0.1 sample, (b)–(f) the distribution image of each element.

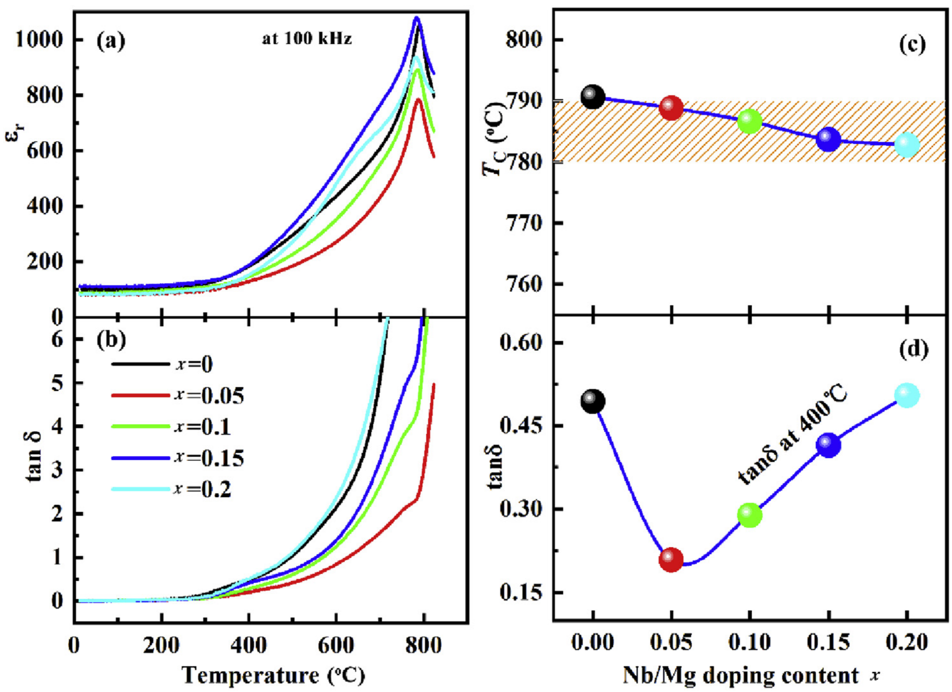

Fig. 5. (a)–(b) Dielectric constant ε_r and dielectric loss tanδ properties of CNM-x samples as a function of temperature, (c) T_C versus Nb–Mg content, (d) tanδ at 400 °C dependence of x.

Temperature dependence of relative dielectric constant (ε_r) and dielectric loss (tanδ) of the CNM-x ceramics measured at 100 kHz are shown in Fig. 5(a) and (b). A dielectric anomalous weak peak can be observed at temperature around 500 °C, which might be attributed to the defect related dielectric response [24,25]. The pristine CBT ceramic displays a Curie temperature (T_C) value of 791 °C, which is in accordance with that reported by Yan et al. [26]. With the increasing Nb–Mg doping level, the T_C gradually decreases to 783 °C. As known, the Curie temperature of BLSFs has a bearing on tolerance factor (t) [27]. The relationship could be expressed as the formula: $t = \frac{R_A + R_O}{\sqrt{2}(R_B + R_O)}$, and the R_A, R_B and R_O inside are the ion radius of A-site cation, B-site cation and oxygen anion respectively [28]. The replacement of Ti^{4+} with larger ions Nb^{5+} and Mg^{2+} would normally lead to a higher Curie temperature due to the decrease of t [29]. The apparent disagreed variation of T_C might be due to the fact that the nd^0 configuration (4d) of Nb^{5+} is higher than that of Ti^{4+} (3d) [24]. Nonetheless, the introduction of Nb–Mg do not cause significant deterioration in T_C. The T_C of CNM-0.1 is still 787 °C, which is close to the pristine CBT (791 °C). The obtained tanδ values at 400 °C are illustrated in Fig. 5(d) as a function of doping concentration. As shown in the figure, the tanδ value decreases firstly, and then increases with the increase of Nb–Mg concentration. It is suggested that suitable amount of Nb–Mg doping could effectively decrease the dielectric loss, which might be ascribed to the reduction of oxygen vacancy concentration. At 400 °C, the dielectric loss of CNM-0.1 ceramic is about 28.9%, which is almost half of that of pristine CBT (49.4%). High Curie temperature along with low dielectric loss indicates that CNM-0.1 might be used under high temperature conditions.

In Fig. 6, the piezoelectric coefficient d_{33} of all ceramics are displayed as a function of Nb–Mg doping level. The pristine CBT ceramic shows a d_{33} value of 10 pC/N, which agrees well with previous reports [17]. All the Nb–Mg co-doped ceramics exhibit higher d_{33} values than the pristine CBT ceramic, revealing that the introduction of Nb–Mg promotes the piezoelectric activity of CBT ceramic evidently. A maximum d_{33} of 24 pC/N is acquired in the composition x = 0.1. A comparison of d_{33} and T_C, including the results obtained in this work and other CBT-based piezoelectric ceramics reported by literatures, is given in Fig. 6(b) [10,16,22,30–37]. Generally, the increase of d_{33} is usually

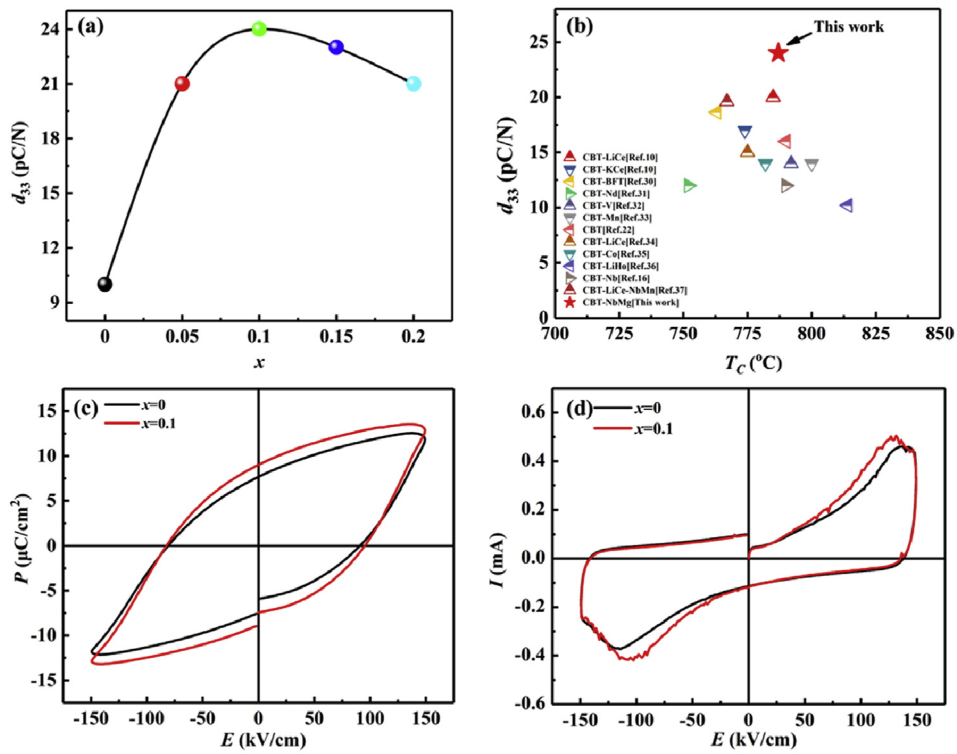

Fig. 6. (a) Variation of piezoelectric coefficient d_{33} with Nb–Mg doping concentration, (b) a comparison of T_C and d_{33} reported by other literatures and present work, (c)*P-E* curves of pure CBT and CNM-0.1 ceramics, (d) *I-E* curves for $x = 0$ and 0.1.

accompanied by a deterioration in T_C, and vice versa. It is obvious that the CNM-0.1 ceramic exhibits a relatively high d_{33} along with a relatively high T_C, implying its great potentials in high temperature applications. Enhancement of piezoelectricity is always related to the improvement of ferroelectricity. Fig. 6 (c) and (d) reveal the ferroelectric hysteresis loops and corresponding switched current densities under 150 kV/mm, at 100 °C and 10 Hz, obtained for CNM-0.1 and pristine CBT. As shown, a slim hysteresis loop could be observed for the pristine CBT ceramic, while a more saturated hysteresis loop is obtained for CNM-0.1 ceramic. Furthermore, more apparent ferroelectric switching current peaks could be observed in CNM-0.1 ceramic, indicating more domain switching during the poling process. In general, the optimized ferroelectric properties should be partly responsible for the enhanced piezoelectric coefficient d_{33} in CNM-0.1 ceramic.

Fig. 7(a) presents the thermal annealing behaviors of d_{33} for all CNM-*x* ceramics ($x = 0$, 0.05, 0.1, 0.15, 0.2). At high temperature, thermal stability is a very important parameter that greatly affects the stability of piezoelectric ceramic. Apparently, the d_{33} value of all the Nb–Mg doped ceramics nearly keeps constant when the annealing temperature is below 500 °C. Above that, it decreases slowly with an increase in annealing temperature, indicating domains begin to reverse and depolarization occurs above this temperature. The d_{33} value decreases sharply when its annealing temperature is closed to the Curie temperature. Still, the CNM-0.1 ceramic exhibits a higher d_{33} of 22 pC/N after annealing at 600 °C, maintaining over 91% of its initial value at room temperature. To give a clear illustration of the obtained thermal stability, a comparison of d_{33} and d_{33T}/d_{33RT} after annealing at 600 °C obtained in this work and other CBT-based piezoelectric ceramics reported by literatures, is summarized in Fig. 7(b) [16,34,36–40]. It is obvious that the CNM-0.1 ceramic possesses an outstanding piezoelectric thermal stability with a significant d_{33}, which makes it more potential for high-temperature piezoelectric applications.

DC resistivity of piezoelectric ceramic is also an important advantage that highly affects the stability of piezoelectric devices at high

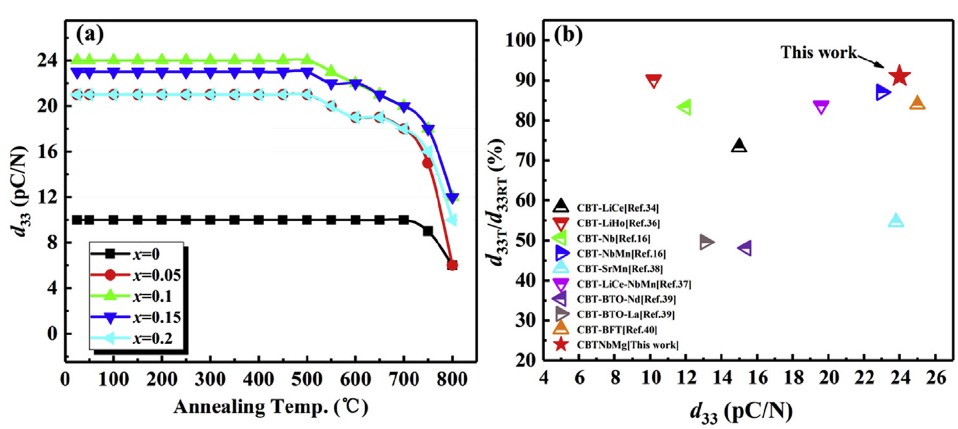

Fig. 7. (a) Piezoelectric coefficient d_{33} CNM-*x* ceramics at various annealing temperature. (b) A comparison of CBT-based ceramics data (T_C and d_{33}) reported by other literatures and present work.

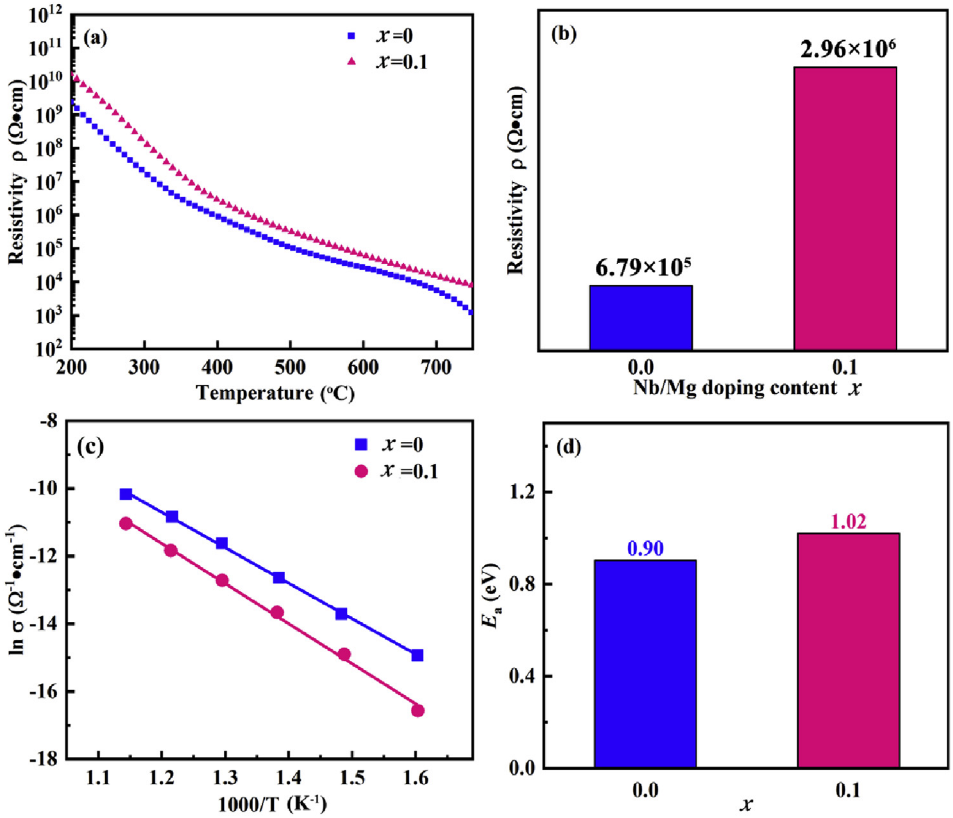

Fig. 8. (a) The DC resistivity ρ of ceramics sample at various temperature, (b) resistivity at 400 °C for ceramics with x = 0, 0.1, (c) Arrhenius for ceramics between conductivity and temperature at x = 0, 0.1, (d) high temperature activation energy for CNM-0.1 and pure CBT.

temperature. Fig. 8(a) displays the DC resistivity ρ of CNM-0.1 and pristine CBT as a function of temperature. As shown, the resistivity ρ of CNM-0.1 ceramic is higher than that of pristine CBT in the entire measurement temperature range. A high resistivity value of 2.96×10^6 Ω•cm is obtained for CNM-0.1 ceramic at 400 °C, which is about 4 times higher than the 6.79×10^5 Ω•cm of pristine CBT ceramic, as shown in Fig. 8(b). In order uncover the related mechanism for the enhanced resistivity, the activation energy E_a is calculated via Arrhenius law:

$$\sigma = \sigma_0 \exp\left(\frac{-E_a}{kT}\right) \tag{1}$$

where σ is conductivity (Ω^{-1}•cm^{-1}), σ_0 is pre exponential factor and E_a is activation energy (eV). The K is the Boltzmann constant (J/K) and T is absolute temperature (K). The Arrhenius curve of σ against $1/T$ of the CNM-0.1 and pristine CBT are shown in Fig. 8(c). The obtained E_a values of pristine CBT and CNM-0.1 ceramics are 0.90 eV, 1.02 eV respectively, which are closed to the threshold activation energy value (1 eV) of oxygen vacancies in BLSFs [41]. An increase in activation energy indicates the motivation of oxygen vacancies becomes difficult, which might be responsible for the increased resistivity. Furthermore, it also implies a lower oxygen vacancies concentration in the Nb–Mg co-doped CBT ceramic [42].

As known, bismuth can easily volatilize during the high temperature preparation process, which would generate the bismuth vacancy [43,44]. To keep the charge neutral, a certain amount of oxygen vacancy will be produced, which might be expressed as Eq. (2):

$$Bi_2O_3 = 2Bi(gas) \uparrow + \tfrac{3}{2}O_2(\text{gas}) \uparrow + 2V_{Bi}''' + 3V_{\ddot{O}} \tag{2}$$

$$\frac{1}{2}O_2 + V_{\ddot{o}} \rightarrow O_o^{\times} + 2h^{\cdot} \tag{3}$$

$$Nb_2O_5 \xrightarrow{2TiO_2} 2Nb_{\dot{Ti}} + 2e' \tag{4}$$

$$Nb_2O_5 + V_{\ddot{O}} \xrightarrow{2TiO_2} 2Nb_{\dot{Ti}} + 5O_O^{\times} \tag{5}$$

$$h^{\cdot} + e' = null \tag{6}$$

$$MgO \xrightarrow{TiO_2} Mg_{Ti}'' + V_{\ddot{O}e} \tag{7}$$

$$Mg_{Ti}'' + V_{\ddot{O}e} \rightarrow (Mg_{Ti}'' - V_{\ddot{O}e}) \tag{8}$$

During the cooling, the oxygen enters into the lattices and generates the hole (h•)(Eq. (3)). The donor doping of Ti^{4+} with Nb^{5+} generates two $2Nb_{\dot{Ti}}$ and two free electrons expressed by Eq. (4), which could eliminate the oxygen vacancies (Eq. (5)) and neutralize the holes in use of electrons (Eq. (6)). On the other hand, the acceptor doping of Ti^{4+} with Mg^{2+} produces one Mg_{Ti}'' and one oxygen vacancy, which can be expressed by Eq. (7). Normally, it would lead to an increase in oxygen vacancy concentration. Nevertheless, the opposite results have been reported in Nb–Mg, Nb–Cu, W–Mn doped BIT ceramics [14,15,45]. In those studies, these oxygen vacancies are treated as extrinsic oxygen vacancies ($V_{\ddot{O}e}$) which may not participate the conduction process, as they will be restricted around defect central by forming strong defect dipoles. To maintain the equilibrium concentration of oxygen vacancies in the ceramic, the number of intrinsic oxygen vacancies will be reduced as the excess extrinsic oxygen vacancies increase. In this study, similar defect dipoles ($Mg_{Ti}'' - V_{\ddot{O}e}$) might be formed in Nb–Mg co-doped CBT ceramics (Eq. (8)), which gives rise to a decrease in intrinsic oxygen vacancy concentration [15,46]. Based on above analysis, the enhancement of DC resistivity in CNM-0.1 ceramic could be ascribed to a synergy effect of the donor Nb and acceptor Mg. Furthermore, it should be noted that the reduction of oxygen vacancy concentration might be also an important reason for the decreased dielectric loss and enhanced ferroelectricity, as shown in Figs. 5 and 6.

4. Conclusions

In summary, mixed-valence ions Nb–Mg co-doped $CaBi_4Ti_{4-x}(Nb_{2/3}Mg_{1/3})_xO_{15}$ (x = 0, 0.05, 0.1, 0.15, 0.2) piezoceramics were prepared by conventional solid-state method. Co-doping of Nb–Mg decreased the oxygen concentration and led to increases in ferroelectricity and piezoelectricity of CBT ceramics. A significantly improved piezoelectric properties were obtained with a high d_{33} value of 24 pC/N accompanied by a high T_C of 787 °C. In addition, the resistivity and thermal stability were also enhanced. These optimized electrical properties demonstrated that the Nb–Mg co-doped $CaBi_4Ti_{4-x}(Nb_{2/3}Mg_{1/3})_xO_{15}$ ceramics could be a potential option for high temperature piezoelectric applications. It further implied that mixed valence ions doping at Ti-sites was a doable way to improve the piezoelectricity of CBT ceramics.

Acknowledgments

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